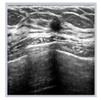
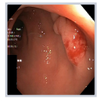
A Multi-domain inputs and task conditioning



ISIC 2018
(Dermoscopy)



BUSI-WHU
(Breast Ultrasound)



Kvasir-SEG
(Endoscopy)



KUB-StoneX
(KUB Radiography)

Task identity / domain label t

Task embedding E(t)

B ReMedFiLM -- FULL Proposed Inference System

ReMedFiLM (G + Bφ + Rψ + Sω)

x

G: Task-conditioned GlobalMedFiLM Backbone

E(t)

Global probability map *Pg*

Global features *Fg*

ROI proposal / candidate region extraction

Cropped candidate ROI *xroi*

Rψ: Regional Residual Refiner

ΔP (corrected local mask)

Bφ: Benefit Router

Sω: Post-Safety Gate

Refine?

SKIP (no useful ROI)

REFINE (beneficial)

Accept?

ACCEPT (use refinement)

REJECT (fallback to Pg)

ROI composition (insert refined local mask)

ReMedFiLM Final Segmentation F

Three legitimate inference outcomes (of ReMedFiLM)

(a) No useful ROI / Skip
→ global output (Pg)

(b) Refine + Accept
→ composed refined output

(c) Refine + Reject
→ global fallback (Pg)

C Staged training strategy

(separate training with progressive dependency)

1 Train G

Train global backbone G (GlobalMedFiLM) on multi-domain data.

2 Generate predictions & ROIs

Generate global predictions (Pg) and ROI crops.

3 Train Rψ

Train regional refiner Rψ on residual correction.

4 Create gain labels

$g = M(P_{refine}) - M(Pg)$

Create paired gain labels / expected refinement benefit (e.g., performance gain g).

5 Train Bφ

Train benefit router Bφ using gain labels.

6 Train Sω

Train post-safety gate Sω (consistency / reliability supervision).

Staged training (G → Rψ → Bφ → Sω)

Inference uses the frozen trained modules: G, Bφ, Rψ and Sω.

Legend

- Inference flow (used in inference)
- Training-only dependency (used in training)
- Global backbone (G: GlobalMedFiLM)
- Task conditioning / ROI module
- Routing / decision module (Bφ, Sω)
- Accepted refinement path (refine + accept, use refined result)
- Rejected fallback path (refine + reject, or direct skip, use Pg)